ASSESSMENT EXERCISE

Multiple Type Questions

1. Which of the following is not a stage in the AI Project Cycle?

(a) Problem Scoping
(b) Data Acquisition
(c) Data Exploration
(d) Prototyping

2. While finalising the aim of this system, you scope the problem which you wish to solve with the help of your project. This is called _____.

(a) Data acquisition
(b) Problem Scoping
(c) Data exploration
(d) Modelling

3. Which of the following would you study in the problem scoping stage of an AI project?

(a) Domain and parameters specific to the problem
(b) Data specific to the problem
(c) Data required for testing
(d) Algorithm and solution specific to the problem

4. What is the role of API (Application Programming Interface) in data acquisition for AI projects?

(a) API is a source of raw visual data
(b) API collects physical data through sensors
(c) API is a messenger that takes requests and provides responses
(d) API is a method of gathering data through surveys

5. It refers to human or mechanical watching, noticing or perceiving of what people actually do or what events take place in a specific working environment.

(a) Survey
(b) Interview
(c) Observation
(d) API

Very Short Answer Type Questions

6. Why is analyzing the current state important in scoping a problem for an AI project?

Ans: Analyzing the current state helps us understand the existing problem, its causes, the environment in which it occurs, and its effects. It helps us identify the right problem and develop an effective AI solution.

7. In the 4Ws Canvas, what does the "Why" block focus on?

Ans: The "Why" block focuses on the importance of solving the problem, the benefits of the solution, and the negative effects of leaving the problem unsolved.

8. What is the first step in scoping a problem in the AI project cycle?

Ans: The first step is to identify the problem that we want to solve using AI and define its aim and scope.

9. What are some techniques used in data exploration?

Ans: Some techniques used in data exploration are:

- Organising data in tables.
- Plotting charts and graphs.
- Creating databases.
- Using data visualization to identify trends and patterns.

Short Answer Type Questions

10. List different sources from where you can collect data.

Ans: Data can be collected from the following sources:

1. Surveys
2. Web scraping
3. Sensors
4. Cameras
5. Observations
6. API (Application Programming Interface)

Government-hosted open-access websites, such as data.gov.in and india.gov.in, are also dependable sources of data.

11. Explain the significance of training data in predictive AI systems.

Ans: Training data is a set of examples used to teach an AI model how to make predictions or perform a task. The model learns patterns from the training data and adjusts its internal parameters accordingly.

For example, an AI model trained with house prices, locations, and sizes can learn patterns to predict the prices of new houses.

12. Describe the role of AI in data exploration.

Ans: AI helps in data exploration by analysing collected data and identifying useful information, patterns, trends, and relationships. It helps us understand large amounts of data and discover insights that support better decision-making.

13. Why is data visualization important in data exploration?

Ans: Data visualization is important because it represents data graphically, making complex information easier to understand. It helps us identify trends, patterns, and relationships in data.

It also helps in comparing information, reducing the complexity of data, and making better decisions. Common visualization techniques include bar charts, pie charts, histograms, scatter plots, and line plots.

Long Answer Type Questions

14. What is a problem scoping stage in the AI project cycle?

Ans: Problem Scoping is the first stage of the AI project cycle. It is the process of identifying a problem that we want to solve using AI and defining the aim and scope of the project.

During this stage, we understand the problem, its causes, the people affected by it, and the importance of solving it. Proper problem scoping helps ensure that the project addresses the correct problem and reduces the chances of project failure.

The 4Ws Problem Canvas is used during problem scoping to understand the problem clearly. It consists of the following four elements:

1. Who: Identifies the stakeholders affected by the problem, directly or indirectly.
2. What: Identifies the main problem or issue that needs to be solved.
3. Where: Identifies the context, location, or environment in which the problem occurs.
4. Why: Explains why solving the problem is important and what benefits the solution will provide.

After completing the 4Ws Canvas, a problem statement is prepared to summarise the main points.

For example, if students have difficulty understanding mathematical concepts, problem scoping helps identify the affected students, the exact difficulty, where it occurs, and why an AI-based learning solution could be useful.

15. What is data exploration, and why is it an essential step in data analysis?

Ans: Data exploration is the third stage of the AI project cycle. It is the process of arranging, examining, and visualising collected data to understand it better.

After data acquisition, the collected data is studied using different techniques to identify its structure, trends, patterns, and relationships. Data can be organised in tables, charts, graphs, or databases.

Importance of data exploration:

1. Identifies patterns and trends: It helps discover whether values are increasing, decreasing, or remaining constant.
2. Improves understanding: It makes large amounts of data easier to interpret.
3. Supports decision-making: It provides useful insights for making informed decisions.
4. Helps select suitable models: It helps determine which method or model is appropriate for further analysis.
5. Makes data easier to understand: Visualization converts numbers into meaningful graphical representations.
6. Reveals relationships: It helps identify relationships between different data features.

Common data visualization techniques include:

- Pie chart
- Bar chart
- Histogram
- Scatter plot
- Line plot
- Timeline
- Area plot

Thus, data exploration is an essential step because it helps us understand the collected data before developing an AI model.